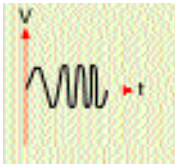


normal value of 60 Hz (the default frequency for conventional AC power in India). This can cause electronic equipment to malfunction and this is especially relevant to computers, as they are so dependent on frequency.



Each of these faults in power can be caused either due to faulty wiring (lack of proper earthing for your power lines), faults in the power lines themselves, errors in the power transmission grid and environmental conditions such as lightning and storms.

...But not powerless

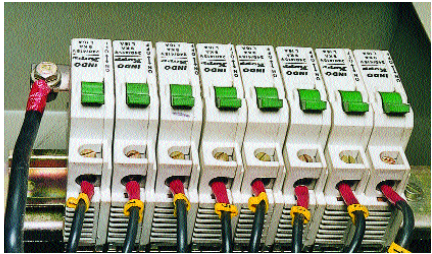
The implications and the extent to which each of these conditions can affect computer hardware are different. Accordingly, there are different solutions available to counter them. The solution you opt for is also decided by your budget and the level of protection you need for your computer hardware.

When combating power problems, there are several weapons with which you can arm your sensitive

electronics components.

■ **Circuit breaker:** This device is a switch capable of responding to dangerous overload levels in an electrical circuit. When installed, the switch detects if the current level of a line is too high and instantaneously (in tens of milliseconds) ‘trips’ and switches off. This effectively disconnects the circuit it is connected to and prevents the equipment on it from receiving dangerously high input power levels. However, since it effectively switches off the power, it’s a very ‘quick and dirty’ way to implement power protection.

Circuit breakers are particularly useful if you want to protect your computer from damage due to failure or malfunction of other equipment that is on the



same circuit as your computer. The circuit breaker does not provide any backup power nor does it regulate the voltage reaching your equipment. For good power protection, this system should be implemented in conjunction with one or more of the other methods stated below.

■ **Spike buster:** This device is almost a standard piece of equipment with most new computer purchases. A spike buster (also called a spike suppressor) looks similar to a power strip that provides added power outlets, except that it contains circuitry that can detect impulse increases in the input voltage (spikes). When it detects a higher voltage, it shuts off and its fuse opens. This prevents a dangerously high voltage from reaching the equipment connected to the spike buster. Such devices can protect your computer in case of errors on the power lines or even lightning strikes, which can cause sharp voltage increases on power lines. However, in case of a power failure, a spike buster will not be able to power your equipment, hence it is still not the best way to implement power protection.

